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Title

Internet BGP peering examples for enterprises

Article Type

Configuration

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Published Date

May 11, 2020

Updated Date

December 11, 2025

Table of Contents

[Internet BGP peering examples for enterprises](#)

[1 Internet BGP peering examples for enterprises](#)

[1.1 Enterprise edge](#)

[1.2 Design choices](#)

[1.2.1 Active/Active or Active/Passive](#)

[1.2.2 Default route or full Internet table](#)

[1.2.3 iBGP/IGP interaction or IGP default route originate](#)

[1.2.4 Route reflector](#)

[1.3 Enterprise BGP configuration examples](#)

[1.3.1 Active/Passive with a full Internet table import](#)

[1.3.2 Active/Passive with default route only from SPs](#)

[1.3.3 Active/Active with full Internet table import](#)

[1.3.4 Active/Active with default route only from SPs](#)

Content

Internet BGP peering examples for enterprises

1 Internet BGP peering examples for enterprises

1.1 Enterprise edge

A typical use case of BGP for an enterprise is to create a possibility to have redundant SPs as Internet uplinks and have its own private IP range (PI) that does not belong to any SP (addresses that belongs to a SP is called PA). This kind of AS is often called a stub AS since it does not have any peers utilizing the network for transit. It is necessary to have control in a stub AS so that traffic from one upstream SP is not forwarded to another. Inside the stub AS iBGP can be used, but also very common is that inside the network an IGP (Interior Gateway Protocol) is used instead. Example: ISIS, OSPF, RIP etc. To interact between BGP and IGP, information can be redistributed so that BGP routing information is made available to the IGP. It is uncommon that IGP routing information is made available to BGP directly since all IGP changes always get reflected to BGP. It is more common to summarize the IP blocks used inside the enterprise AS and make them available to BGP. Another method is to let eBGP routers originate a default route into the IGP to draw the traffic to the edge and statically announce the IP addresses owned by the enterprise.

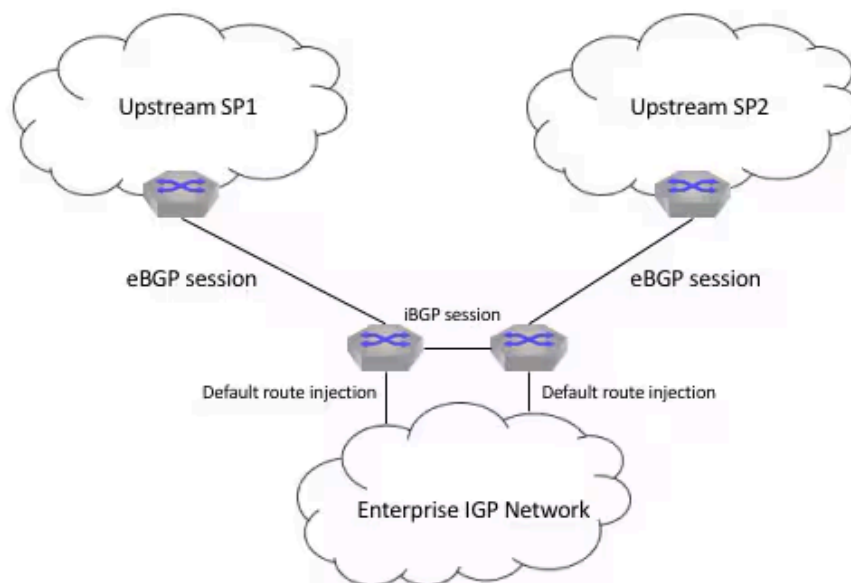


Fig. Enterprise edge BGP use case

1.2 Design choices

There are a few design choices to make in the enterprise edge use case. These choices may already been done if implementing in a brownfield deployment or needs to be done if it is a greenfield deployment.

1.2.1 Active/Active or Active/Passive

Due the nature of BGP and how the decision of which routes to use when having two separate BGP peers, is it very easy to decide whether to use one peer as a primary path or send traffic the best path using all peers. I.e. whether to implement an active/passive or active/active scenario. To implement the chosen strategy, many BGP route attributes can be used. But easiest is to use local preference or weight. Simply set a higher local preference or weight with a route policy when receiving the routes from your preferred peer in an active/passive scenario. In an active/active scenario make sure all peers are having the same incoming route policy. Then the best route for each destination will be used.

Since the decision is made locally in each AS which routes to use, it is harder to control which paths other AS use vs your own AS. The tools available are as path and med (multi exit discriminator). Med is only usable when having two or more peering sessions with same AS neighbor. Use a route policy to set med value when sending prefixes to the neighbor. If the neighbor respects the med value that is announced, the path with lowest med is preferred. When addressing the need to try to mark one path as less preferred when having two or more BGP peering sessions with different AS neighbors, a method called as path prepend can be used. This is implemented through having a route policy that adds your own AS several times

on prefixes announced to an AS neighbor to make that a less preferred path. Since this makes the as path considerably longer on your less preferred path than your preferred, this influences other AS networks to use the shorter path.

1.2.2 Default route or full Internet table

It is possible to choose to only receive a default route via BGP from the neighbors or to receive a full Internet table if the neighbor have that available. In an active/active scenario if a full Internet table is received, the BGP best path algorithm will choose a best path for each received prefix. In an active/active scenario where only default routes are received and attributes are making the routes equal, ECMP will try to distribute the traffic on both links. In an active/passive scenario it is possible to either receive a default route from all neighbors, receive a full Internet table from all neighbors or to receive a default route from

the less preferred neighbors and a full Internet table from the preferred neighbors. Using a route policy setting a higher local preference or weight for preferred neighbors is done in the same way in all these scenarios.

1.2.3 iBGP/IGP interaction or IGP default route originate

In most enterprise edge scenarios all internal networking is taken care of by firewalls facing the Internet edge and internal routers totally separate from the external network. Then this design choice is irrelevant. But in some cases the network is of such a size that external networks is also spread internally so that the Internet edge routers participate in the routing before any firewalls are encountered. In this case this design choice need to be made.

Based on the size of the network either BGP can be used on the Internet edge only and then have an IGP taking care of the routing inside the AS. The BGP Internet edge routers then simply also participate in the IGP and use default information originate to attract traffic. Also iBGP may be used internally in the network carrying the prefixes used for deploying services and workstations and then let the IGP carry prefixes used for connectivity between BGP routers.

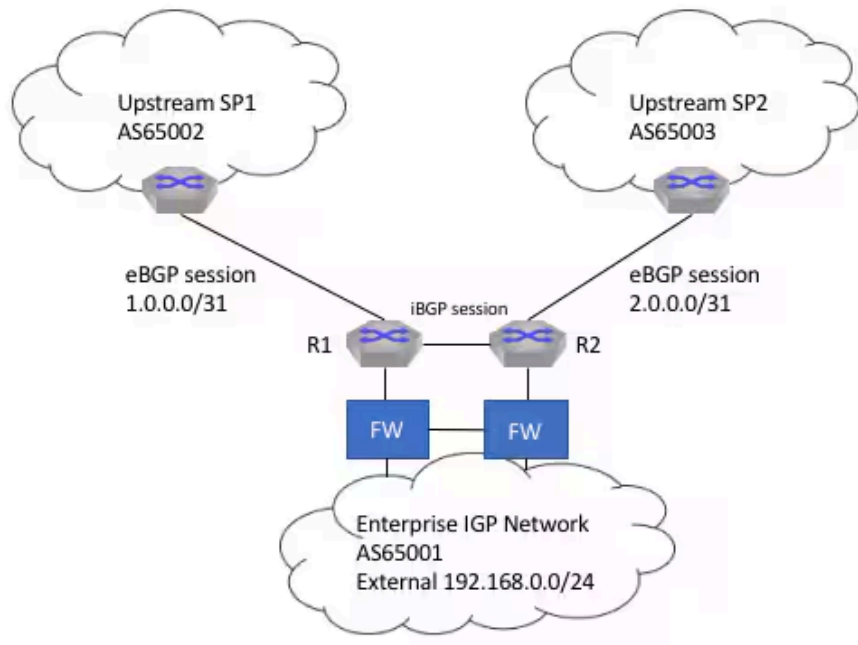
If a default route is used in the IGP in this scenario, it is important that iBGP routers are directly connected or a route loop can occur.

1.2.4 Route reflector

If iBGP is used it needs to be decided whether to deploy a fully meshed iBGP session architecture or if route reflectors should be used. The reason to choose to use route reflector is to simplify the configuration if iBGP peers. Instead of configuring a session for each iBGP router in the network, a session to the route reflectors are configured. In the route reflector a session for each iBGP router is still configured.

1.3 Enterprise BGP configuration examples

Ponder the following network:



1.3.1 Active/Passive with a full Internet table import

Note how R2 sets a lower local preference than R1 with an inbound route-map. And that R2 uses an outbound route-map to AS prepend to make the path less desirable. Also all routes are accepted in R1 and R2 from the Upstream SPs with a BOGON control. The result is that a full Internet table will be received and used.

R1

```
!  
hostname R1  
!  
interface Ethernet1  
  description "Connection to upstream SP1"  
  no switchport  
  ip address 1.0.0.1/31  
!  
interface Ethernet2  
  description "iBGP connection to R2"  
  no switchport  
  ip address 3.0.0.0/31  
!  
interface Ethernet3  
  description "Connection to inside network and FW cluster"  
  no switchport  
  ip address 192.168.0.2/24  
  vrrp 1 ip 192.168.0.1  
!  
interface Loopback0  
  ip address 4.0.0.0/32  
!  
ip as-path access-list LocalAS permit ^$ any  
!  
ip hardware fib optimize prefixes profile internet  
!  
ip prefix-list BOGON-Prefixes seq 10 permit 10.0.0.0/8  
ip prefix-list BOGON-Prefixes seq 20 permit 172.16.0.0/12  
ip prefix-list BOGON-Prefixes seq 30 permit 192.168.0.0/16  
ip prefix-list LocalAS seq 10 permit 192.168.0.0/24  
!  
ip routing  
!  
router bgp 65001  
  router-id 4.0.0.0  
  neighbor iBGPNeighbors peer-group  
  neighbor iBGPNeighbors next-hop-self  
  neighbor iBGPNeighbors remote-as 65001
```

```
neighbor iBGPNeighbors soft-reconfiguration inbound all
neighbor iBGPNeighbors password 0 mysecret
neighbor UpstreamSP1 peer-group
neighbor UpstreamSP1 remote-as 65002
neighbor UpstreamSP1 remove-private-as
neighbor UpstreamSP1 soft-reconfiguration inbound all
neighbor UpstreamSP1 idle-restart-timer 60
neighbor UpstreamSP1 route-map InboundSP1 in
neighbor UpstreamSP1 route-map OutboundSP1 out
neighbor UpstreamSP1 password 0 mysecret
neighbor 1.0.0.0 peer-group UpstreamSP1
neighbor 3.0.0.1 peer-group iBGPNeighbors
network 192.168.0.0/24
!
route-map BOGONS permit 10
  match ip address prefix-list BOGON-Prefixes
!
route-map BOGONS deny 20
!
route-map InboundSP1 deny 10
  sub-route-map BOGONS
!
route-map InboundSP1 permit 20
  set local-preference 200
!
route-map OutboundSP1 permit 10
  match as-path LocalAS
  match ip address prefix-list LocalAS
  set origin igp
!
route-map OutboundSP1 deny 20
!
```

R2

```
!  
hostname R2  
!  
interface Ethernet1  
  description "Connection to upstream SP2"  
  no switchport  
  ip address 2.0.0.1/31  
!  
interface Ethernet2  
  description "iBGP connection to R2"  
  no switchport  
  ip address 3.0.0.1/31  
!  
interface Ethernet3  
  description "Connection to inside network and FW cluster"  
  no switchport  
  ip address 192.168.0.3/24  
  vrrp 1 ip 192.168.0.1  
!  
interface Loopback0  
  ip address 4.0.0.1/32  
!  
ip as-path access-list LocalAS permit ^$ any  
!  
ip hardware fib optimize prefixes profile internet  
!  
ip prefix-list BOGON-Prefixes seq 10 permit 10.0.0.0/8  
ip prefix-list BOGON-Prefixes seq 20 permit 172.16.0.0/12  
ip prefix-list BOGON-Prefixes seq 30 permit 192.168.0.0/16  
ip prefix-list LocalAS seq 10 permit 192.168.0.0/24  
!  
ip routing  
!  
router bgp 65001  
  router-id 4.0.0.1  
  neighbor iBGPNeighbors peer-group  
  neighbor iBGPNeighbors next-hop-self  
  neighbor iBGPNeighbors remote-as 65001
```



```
neighbor iBGPNeighbors soft-reconfiguration inbound all
neighbor iBGPNeighbors password 0 mysecret
neighbor UpstreamSP2 peer-group
neighbor UpstreamSP2 remote-as 65002
neighbor UpstreamSP2 remove-private-as
neighbor UpstreamSP2 soft-reconfiguration inbound all
neighbor UpstreamSP2 idle-restart-timer 60
neighbor UpstreamSP2 route-map InboundSP2 in
neighbor UpstreamSP2 route-map OutboundSP2 out
neighbor UpstreamSP2 password 0 mysecret
neighbor 2.0.0.0 peer-group UpstreamSP2
neighbor 3.0.0.0 peer-group iBGPNeighbors
network 192.168.0.0/24
!
route-map BOGONS permit 10
  match ip address prefix-list BOGON-Prefixes
!
route-map BOGONS deny 20
!
route-map InboundSP2 deny 10
  sub-route-map BOGONS
!
route-map InboundSP2 permit 20
  set local-preference 100
!
route-map OutboundSP2 permit 10
  match as-path LocalAS
  match ip address prefix-list LocalAS
  set as-path prepend auto
  set origin igp
!
route-map OutboundSP2 deny 20
!
```

1.3.2 Active/Passive with default route only from SPs

Here both routers accept the default route only from Upstream SPs and set different local preferences so that the network will prefer R1 as path to Internet. R2 also prepends via a route-map to make the path via R2 to the local AS less desirable for other AS neighbors.

To protect against excessive announcements from SPs, maximum-routes is used. Since explicit control is done on which prefixes that are received, no BOGON control is needed.

R1

```
!  
hostname R1  
!  
interface Ethernet1  
  description "Connection to upstream SP1"  
  no switchport  
  ip address 1.0.0.1/31  
!  
interface Ethernet2  
  description "iBGP connection to R2"  
  no switchport  
  ip address 3.0.0.0/31  
!  
interface Ethernet3  
  description "Connection to inside network and FW cluster"  
  no switchport  
  ip address 192.168.0.2/24  
  vrrp 1 ip 192.168.0.1  
!  
interface Loopback0  
  ip address 4.0.0.0/32  
!  
ip as-path access-list LocalAS permit ^$ any  
!  
ip hardware fib optimize prefixes profile internet  
!  
ip prefix-list LocalAS seq 10 permit 192.168.0.0/24  
ip prefix-list DefaultRoute seq 10 permit 0.0.0.0/0  
!  
ip routing  
!  
router bgp 65001  
  router-id 4.0.0.0  
  neighbor iBGPNeighbors peer-group  
  neighbor iBGPNeighbors next-hop-self  
  neighbor iBGPNeighbors remote-as 65001  
  neighbor iBGPNeighbors soft-reconfiguration inbound all  
  neighbor iBGPNeighbors password 0 mysecret
```

```
neighbor UpstreamSP1 peer-group
neighbor UpstreamSP1 remote-as 65002
neighbor UpstreamSP1 remove-private-as
neighbor UpstreamSP1 soft-reconfiguration inbound all
neighbor UpstreamSP1 idle-restart-timer 60
neighbor UpstreamSP1 route-map InboundSP1 in
neighbor UpstreamSP1 route-map OutboundSP1 out
neighbor UpstreamSP1 password 0 mysecret
neighbor UpstreamSP1 maximum-routes 10
neighbor 1.0.0.0 peer-group UpstreamSP1
neighbor 3.0.0.1 peer-group iBGPNeighbors
network 192.168.0.0/24
!
route-map InboundSP1 permit 10
  match ip address prefix-list DefaultRoute
  set local-preference 200
!
route-map InboundSP1 deny 20
!
route-map OutboundSP1 permit 10
  match as-path LocalAS
  match ip address prefix-list LocalAS
  set origin igp
!
route-map OutboundSP1 deny 20
!
```

R2

```
!  
hostname R2  
!  
interface Ethernet1  
  description "Connection to upstream SP2"  
  no switchport  
  ip address 2.0.0.1/31  
!  
interface Ethernet2  
  description "iBGP connection to R2"  
  no switchport  
  ip address 3.0.0.1/31  
!  
interface Ethernet3  
  description "Connection to inside network and FW cluster"  
  no switchport  
  ip address 192.168.0.3/24  
  vrrp 1 ip 192.168.0.1  
!  
interface Loopback0  
  ip address 4.0.0.1/32  
!  
ip as-path access-list LocalAS permit ^$ any  
!  
ip hardware fib optimize prefixes profile internet  
!  
ip prefix-list DefaultRoute seq 10 permit 0.0.0.0/0  
ip prefix-list LocalAS seq 10 permit 192.168.0.0/24  
!  
ip routing  
!  
router bgp 65001  
  router-id 4.0.0.1  
  neighbor iBGPNeighbors peer-group  
  neighbor iBGPNeighbors next-hop-self  
  neighbor iBGPNeighbors remote-as 65001  
  neighbor iBGPNeighbors soft-reconfiguration inbound all  
  neighbor iBGPNeighbors password 0 mysecret
```

```
neighbor UpstreamSP2 peer-group
neighbor UpstreamSP2 remote-as 65002
neighbor UpstreamSP2 remove-private-as
neighbor UpstreamSP2 soft-reconfiguration inbound all
neighbor UpstreamSP2 idle-restart-timer 60
neighbor UpstreamSP2 route-map InboundSP2 in
neighbor UpstreamSP2 route-map OutboundSP2 out
neighbor UpstreamSP2 password 0 mysecret
neighbor UpstreamSP1 maximum-routes 10
neighbor 2.0.0.0 peer-group UpstreamSP2
neighbor 3.0.0.0 peer-group iBGPNeighbors
network 192.168.0.0/24
!
route-map InboundSP2 permit 10
  match ip address prefix-list DefaultRoute
  set local-preference 100
!
route-map InboundSP2 deny 20
!
route-map OutboundSP2 permit 10
  match as-path LocalAS
  match ip address prefix-list LocalAS
  set as-path prepend auto
  set origin igp
!
route-map OutboundSP2 deny 20
!
```

1.3.3 Active/Active with full Internet table import

The Internet edge routers will treat the prefixes the same and exchange them with each other with iBGP. Best path (shortest as-path) will be selected. Also EOS feature with relaxed as path control for multipath is used to try to use ECMP for paths with equal as-path. If prefix to be used exists in the iBGP neighboring router, the connection between R1 and R2 will be used before continuing out to the upstream SP. Local AS prefix is announced upstream with equal attributes to allow neighboring AS to choose the path they prefer.

R1

```
!  
hostname R1  
!  
interface Ethernet1  
  description "Connection to upstream SP1"  
  no switchport  
  ip address 1.0.0.1/31  
!  
interface Ethernet2  
  description "iBGP connection to R2"  
  no switchport  
  ip address 3.0.0.0/31  
!  
interface Ethernet3  
  description "Connection to inside network and FW cluster"  
  no switchport  
  ip address 192.168.0.2/24  
  vrrp 1 ip 192.168.0.1  
!  
interface Loopback0  
  ip address 4.0.0.0/32  
!  
ip as-path access-list LocalAS permit ^$ any  
!  
ip hardware fib optimize prefixes profile internet  
!  
ip prefix-list LocalAS seq 10 permit 192.168.0.0/24  
!  
ip routing  
!  
router bgp 65001  
  router-id 4.0.0.0  
  maximum-paths 2 ecmp 2  
  neighbor iBGPNeighbors peer-group  
  neighbor iBGPNeighbors next-hop-self  
  neighbor iBGPNeighbors remote-as 65001  
  neighbor iBGPNeighbors soft-reconfiguration inbound all  
  neighbor iBGPNeighbors password 0 mysecret
```

```
neighbor UpstreamSP1 peer-group
neighbor UpstreamSP1 remote-as 65002
neighbor UpstreamSP1 remove-private-as
neighbor UpstreamSP1 soft-reconfiguration inbound all
neighbor UpstreamSP1 idle-restart-timer 60
neighbor UpstreamSP1 route-map InboundSP1 in
neighbor UpstreamSP1 route-map OutboundSP1 out
neighbor UpstreamSP1 password 0 mysecret
neighbor 1.0.0.0 peer-group UpstreamSP1
neighbor 3.0.0.1 peer-group iBGPNeighbors
network 192.168.0.0/24
!
route-map InboundSP1 permit 10
  set local-preference 200
!
route-map OutboundSP1 permit 10
  match as-path LocalAS
  match ip address prefix-list LocalAS
  set origin igp
!
route-map OutboundSP1 deny 20
!
```

R2


```
!  
hostname R2  
!  
interface Ethernet1  
  description "Connection to upstream SP2"  
  no switchport  
  ip address 2.0.0.1/31  
!  
interface Ethernet2  
  description "iBGP connection to R2"  
  no switchport  
  ip address 3.0.0.1/31  
!  
interface Ethernet3  
  description "Connection to inside network and FW cluster"  
  no switchport  
  ip address 192.168.0.3/24  
  vrrp 1 ip 192.168.0.1  
!  
interface Loopback0  
  ip address 4.0.0.1/32  
!  
ip as-path access-list LocalAS permit ^$ any  
!  
ip hardware fib optimize prefixes profile internet  
!  
ip prefix-list LocalAS seq 10 permit 192.168.0.0/24  
!  
ip routing  
!  
router bgp 65001  
  router-id 4.0.0.1  
  maximum-paths 2 ecmp 2  
  neighbor iBGPNeighbors peer-group  
  neighbor iBGPNeighbors remote-as 65001  
  neighbor iBGPNeighbors next-hop-self  
  neighbor iBGPNeighbors soft-reconfiguration inbound all  
  neighbor iBGPNeighbors password 0 mysecret
```

```
neighbor UpstreamSP2 peer-group
neighbor UpstreamSP2 remote-as 65002
neighbor UpstreamSP2 remove-private-as
neighbor UpstreamSP2 soft-reconfiguration inbound all
neighbor UpstreamSP2 idle-restart-timer 60
neighbor UpstreamSP2 route-map InboundSP2 in
neighbor UpstreamSP2 route-map OutboundSP2 out
neighbor UpstreamSP2 password 0 mysecret
neighbor 2.0.0.0 peer-group UpstreamSP2
neighbor 3.0.0.0 peer-group iBGPNeighbors
network 192.168.0.0/24
!
route-map InboundSP2 permit 10
  set local-preference 200
!
route-map OutboundSP2 permit 10
  match as-path LocalAS
  match ip address prefix-list LocalAS
  set origin igp
!
route-map OutboundSP2 deny 20
!
```

1.3.4 Active/Active with default route only from SPs

Here both routers accept only the default route from upstream SPs and set same local preference. To use both default routes, maximum-paths is added to allow ECMP. Local AS prefix is announced upstream with equal attributes to allow neighboring AS to choose the path they prefer. Since exact control on what prefixes are received from upstream SPs, no BOGON control is needed. To protect against excessive announcements from upstream SPs, maximum-routes is used.

R1

```
!  
hostname R1  
!  
interface Ethernet1  
  description "Connection to upstream SP1"  
  no switchport  
  ip address 1.0.0.1/31  
!  
interface Ethernet2  
  description "iBGP connection to R2"  
  no switchport  
  ip address 3.0.0.0/31  
!  
interface Ethernet3  
  description "Connection to inside network and FW cluster"  
  no switchport  
  ip address 192.168.0.2/24  
  vrrp 1 ip 192.168.0.1  
!  
interface Loopback0  
  ip address 4.0.0.0/32  
!  
ip as-path access-list LocalAS permit ^$ any  
!  
ip hardware fib optimize prefixes profile internet  
!  
ip prefix-list LocalAS seq 10 permit 192.168.0.0/24  
ip prefix-list DefaultRoute seq 10 permit 0.0.0.0/0  
!  
ip routing  
!  
router bgp 65001  
  router-id 4.0.0.0  
  maximum-paths 2 ecmp 2  
  neighbor iBGPNeighbors peer-group  
  neighbor iBGPNeighbors next-hop-self  
  neighbor iBGPNeighbors remote-as 65001  
  neighbor iBGPNeighbors soft-reconfiguration inbound all
```

```
neighbor iBGPNeighbors password 0 mysecret
neighbor UpstreamSP1 peer-group
neighbor UpstreamSP1 remote-as 65002
neighbor UpstreamSP1 remove-private-as
neighbor UpstreamSP1 soft-reconfiguration inbound all
neighbor UpstreamSP1 idle-restart-timer 60
neighbor UpstreamSP1 route-map InboundSP1 in
neighbor UpstreamSP1 route-map OutboundSP1 out
neighbor UpstreamSP1 password 0 mysecret
Neighbor UpstreamSP1 maximum-routes 10
neighbor 1.0.0.0 peer-group UpstreamSP1
neighbor 3.0.0.1 peer-group iBGPNeighbors
network 192.168.0.0/24
!
route-map InboundSP1 permit 10
  match ip address prefix-list DefaultRoute
  set local-preference 200
!
route-map InboundSP1 deny 20
!
route-map OutboundSP1 permit 10
  match as-path LocalAS
  match ip address prefix-list LocalAS
  set origin igp
!
route-map OutboundSP1 deny 20
!
```

R2

```
!  
hostname R2  
!  
interface Ethernet1  
  description "Connection to upstream SP2"  
  no switchport  
  ip address 2.0.0.1/31  
!  
interface Ethernet2  
  description "iBGP connection to R2"  
  no switchport  
  ip address 3.0.0.1/31  
!  
interface Ethernet3  
  description "Connection to inside network and FW cluster"  
  no switchport  
  ip address 192.168.0.3/24  
  vrrp 1 ip 192.168.0.1  
!  
interface Loopback0  
  ip address 4.0.0.1/32  
!  
ip as-path access-list LocalAS permit ^$ any  
!  
ip hardware fib optimize prefixes profile internet  
!  
ip prefix-list LocalAS seq 10 permit 192.168.0.0/24  
ip prefix-list DefaultRoute seq 10 permit 0.0.0.0/0  
!  
ip routing  
!  
router bgp 65001  
  router-id 4.0.0.1  
  maximum-paths 2 ecmp 2  
  neighbor iBGPNeighbors peer-group  
  neighbor iBGPNeighbors next-hop-self  
  neighbor iBGPNeighbors remote-as 65001  
  neighbor iBGPNeighbors soft-reconfiguration inbound all
```

```
neighbor iBGPNeighbors password 0 mysecret
neighbor UpstreamSP2 peer-group
neighbor UpstreamSP2 remote-as 65002
neighbor UpstreamSP2 remove-private-as
neighbor UpstreamSP2 soft-reconfiguration inbound all
neighbor UpstreamSP2 idle-restart-timer 60
neighbor UpstreamSP2 route-map InboundSP2 in
neighbor UpstreamSP2 route-map OutboundSP2 out
neighbor UpstreamSP2 password 0 mysecret
neighbor UpstreamSP2 maximum-routes 10
neighbor 2.0.0.0 peer-group UpstreamSP2
neighbor 3.0.0.0 peer-group iBGPNeighbors
network 192.168.0.0/24
!
route-map InboundSP2 permit 10
  match ip address prefix-list DefaultRoute
  set local-preference 200
!
route-map InboundSP2 deny 20
!
route-map OutboundSP2 permit 10
  match as-path LocalAS
  match ip address prefix-list LocalAS
  set origin igp
!
route-map OutboundSP2 deny 20
!
```

Enterprises seeking redundancy for their Internet connectivity and agility to change service provider when needed, greatly benefit from having their own AS number and IP addresses that can be announced using their own BGP routers. This article features some design considerations and configuration to achieve this in a common enterprise scenario.

Routing/Switching
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